

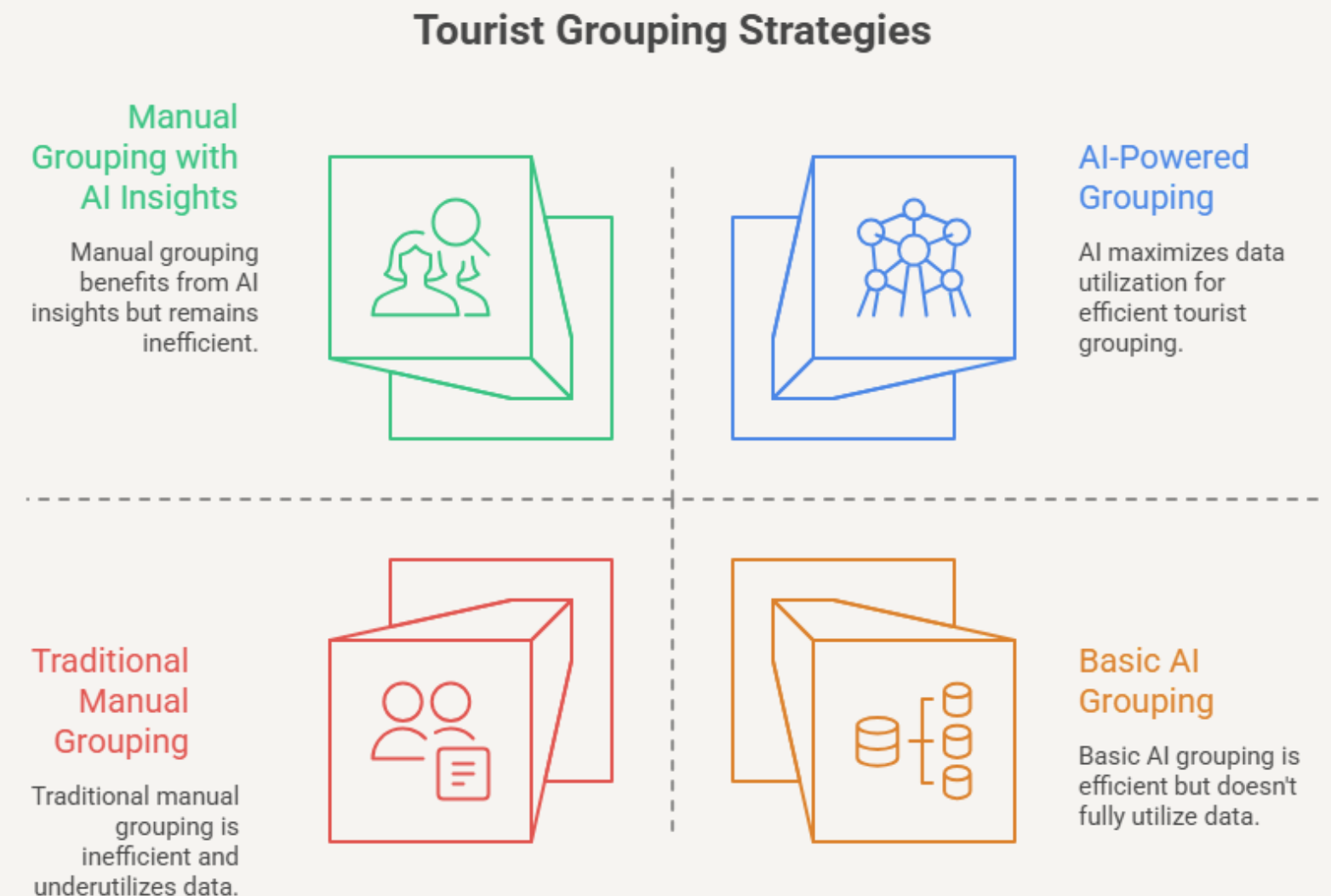
Tourist Clustering



Presented by Nissan Shrestha
London Met ID : 23048990

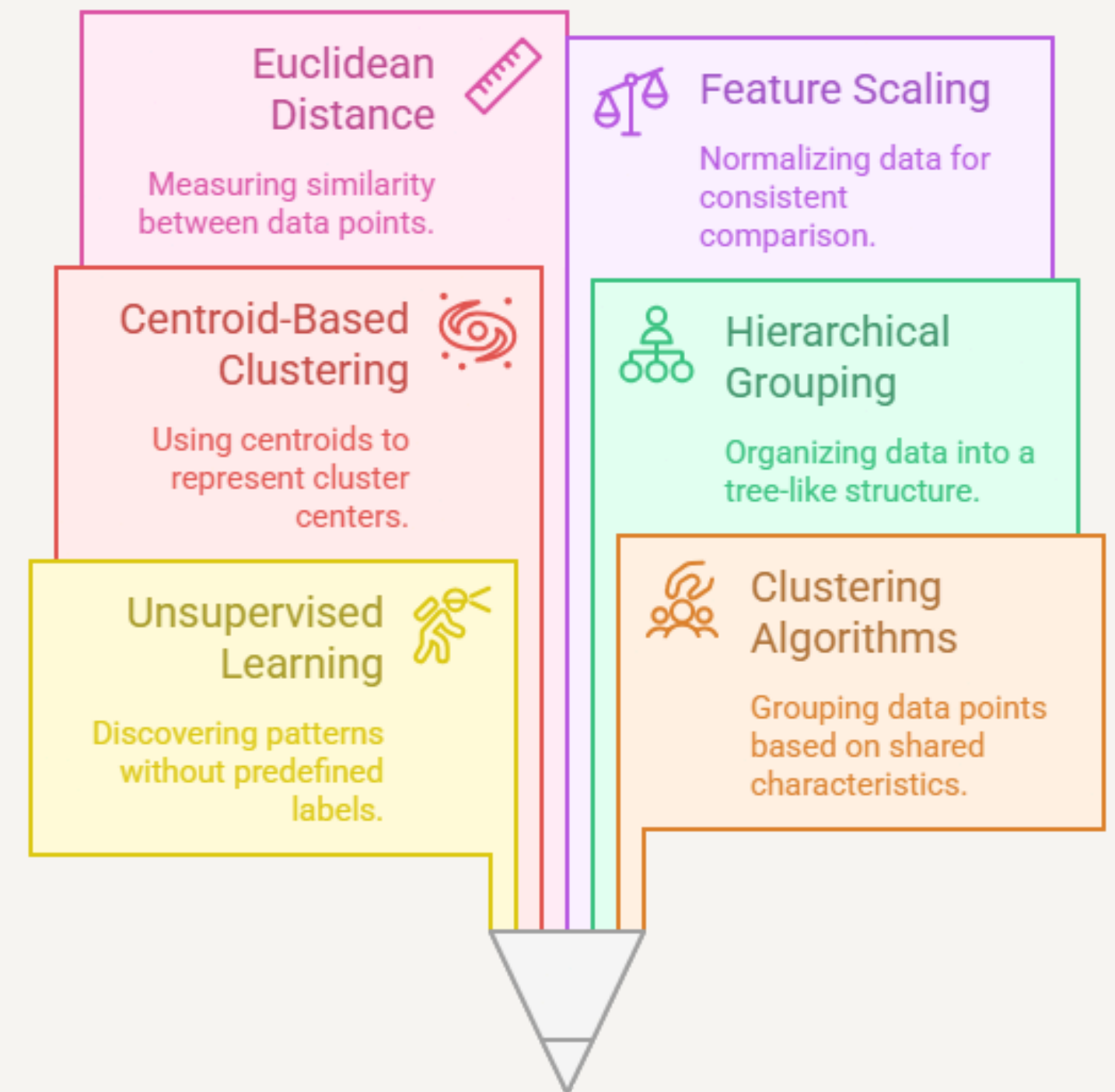
Introduction

- Tourism platforms collect large volumes of tourist preference data
- Tourists have diverse interests in attractions and activities
- Manual grouping of tourists is time-consuming and inefficient
- Artificial Intelligence helps identify hidden patterns in data
- The project focuses on grouping tourists based on similar interests



Concepts Used

- Unsupervised machine learning for pattern discovery
- Clustering algorithms for data segmentation
- Centroid-based clustering concepts
- Hierarchical grouping of data points based on similarity
- Euclidean distance for similarity measurement
- Feature scaling using Min-Max normalization



Background

1. “Evaluation of Partitioning Clustering Algorithms for Processing Social Media Data in Tourism Domain” (Renjith et al., 2018)

- Shows clustering is effective for tourist preference analysis
- Supports using TripAdvisor reviews for segmentation
- Confirms K-Means is commonly used in tourism data

2. “A Review of Data-Driven Market Segmentation in Tourism” (Dolnicar, 2002)

- Highlights limitations of traditional demographic segmentation
- Supports data-driven clustering for better tourism targeting
- Justifies why clustering is useful for tourism marketing

3. “Role of Social Media in Online Travel Information Search” (Xiang & Gretzel, 2010)

- Confirms social media reviews provide rich tourist preference data
- Shows user-generated reviews are valuable for analyzing behavior
- Supports using online review data for clustering tourists

Dataset Used

UC Irvine
Machine Learning
Repository

[Datasets](#)[Contribute Dataset](#)[About Us](#)

[Login](#)



Travel Reviews

Donated on 12/18/2018

Reviews on destinations in 10 categories mentioned across East Asia. Each traveler rating is mapped as Excellent(4), Very Good(3), Average(2), Poor(1), and Terrible(0) and average rating is used.

Dataset Characteristics	Subject Area	Associated Tasks
Multivariate, Text	Other	Classification, Clustering
Feature Type	# Instances	# Features
Real	980	10

Dataset Information

Additional Information
This data set is populated by crawling TripAdvisor.com. Reviews on destinations in 10 categories mentioned across East Asia are considered. Each traveler rating is mapped as Excellent (4), Very Good (3), Average (2), Poor (1), and Terrible (0) and average rating is used against each category per user.

Has Missing Values?
No

[Download](#) (56.1 KB)

 [Import in Python](#)

[Cite](#)

👤 1 citations

👁 33700 views

Creators

 Shini Renjith

DOI

10.24432/C56K6W

License

This dataset is licensed under a [Creative Commons Attribution 4.0 International](#) (CC BY 4.0) license.

Proposed Solution

Tourist Clustering System Overview

- Segment tourists based on their preferences for different destination types
- Use unsupervised learning to discover natural tourist groups
- Create meaningful cluster profiles (labels) for each group

Key Steps in the Solution

1. Data Preprocessing

- Remove user ID column
- Normalize ratings using Min-Max scaling

2. Clustering

- Apply K-Means to generate clusters
- Apply Hierarchical Clustering for comparison

Key Steps in the Solution

3. Cluster Interpretation

- Calculate mean ratings per cluster
- Assign descriptive labels (e.g., “Nature & Beach tourists”)

4. Application

- CLI menu to view cluster summaries and classify new user

Outcome

Generate tourist segments that can be used for:

- Personalized recommendations
- Marketing strategies
- Travel planning

Pseudocode

INPUT: Dataset D of n tourists with 10 rating features

OUTPUT: Two clustering solutions and cluster labels

BEGIN

1. DATA PREPARATION

- LOAD dataset

- RENAME columns to destination names

- DROP 'User ID' column

- CHECK for missing values

- APPLY Min-Max SCALING to all rating features

Pseudocode

2.K-MEANS CLUSTERING

DEFINE K_RANGE from 2 to 10

FOR each K in K_RANGE DO

 RUN K-MEANS with K clusters

 CALCULATE WCSS (Elbow method)

 CALCULATE SILHOUETTE SCORE

END FOR

SELECT optimal K using elbow + silhouette

RUN FINAL K-MEANS with chosen K (K=6)

STORE cluster labels

CALCULATE cluster MEANS for each feature

ASSIGN descriptive LABELS to clusters

Pseudocode

3. HIERARCHICAL CLUSTERING

RUN AGGLOMERATIVE CLUSTERING (Ward linkage)

GENERATE DENDROGRAM for visual analysis

CUT dendrogram to create 6 clusters

ASSIGN cluster labels to dataset

CALCULATE cluster MEANS for each feature

ASSIGN descriptive LABELS to clusters

4. COMPARISON

CREATE CROSS-TABULATION between K-MEANS and HIERARCHICAL clusters

VISUALIZE using HEATMAP

Pseudocode

5.APPLICATION (CLI)

DISPLAY menu options:

1. VIEW K-MEANS summary
2. VIEW HIERARCHICAL summary
3. COMPARE both
4. ASSIGN cluster to new user
5. EXIT

IF user selects option 4 THEN

ACCEPT ratings input (0 to 4)

SCALE input using SAME SCALER

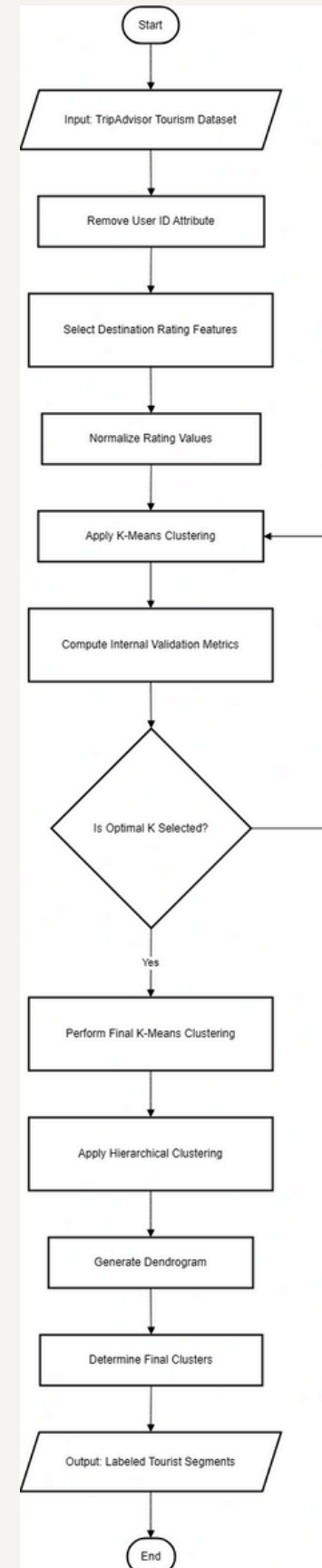
PREDICT cluster using K-MEANS model

DISPLAY predicted cluster and label

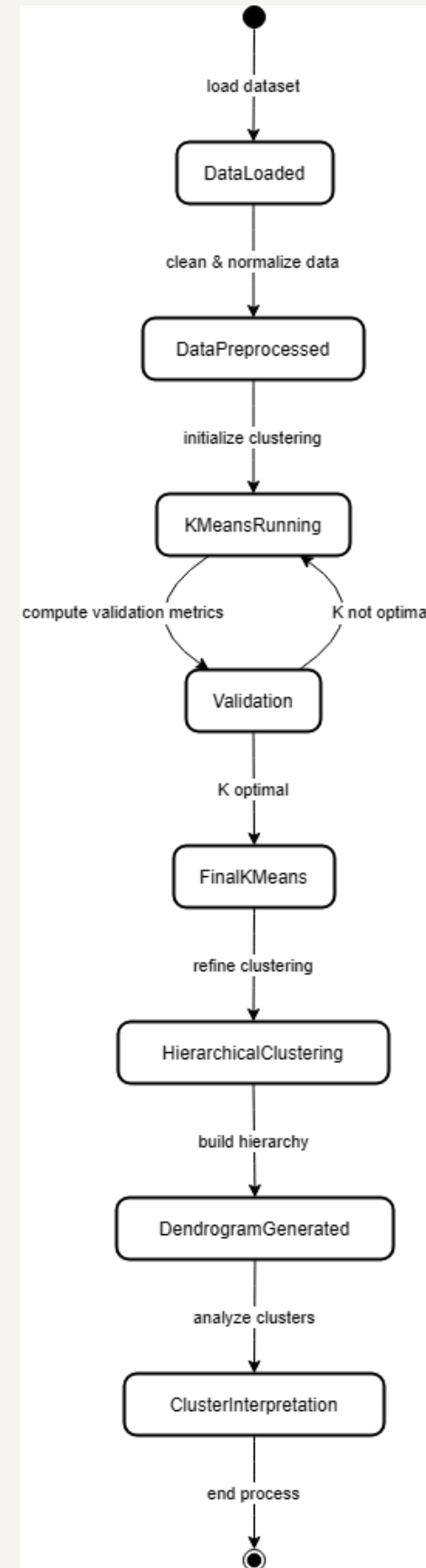
END IF

END

Flowchart



State Transition Diagram



Application (CLI)

```
: main()
```

```
=== Tourist Clustering Application ===
```

1. View K-Means Cluster Summary
2. View Hierarchical Cluster Summary
3. Compare K-Means vs Hierarchical
4. Assign cluster for a new user (K-Means)
5. Exit

```
Enter choice (1-5): 
```

Application (CLI)

```
Enter choice (1-5): 1
      Art_Galleries  Dance_Clubs  Juice_Bars  Restaurants  Museums  \
Cluster
0      0.862044      1.208467      0.579197      0.510803      0.970365
1      0.872330      1.459773      2.259773      0.595568      1.131250
2      0.933797      1.332193      0.418021      0.386845      0.755508
3      0.839908      1.329217      0.390968      0.593687      0.673088
4      0.970272      1.527075      1.174150      0.649796      1.356599
5      0.898190      1.215862      1.554828      0.434138      0.880517

      Resorts      Parks      Beaches      Theaters      Religious_Institutions
Cluster
0      1.953723      3.179781      2.996642      1.992117      2.844161
1      2.170341      3.193239      2.791477      1.585057      2.462500
2      1.434011      3.170963      2.838770      1.463957      3.256043
3      1.582120      3.179631      2.812396      1.542304      2.714700
4      2.421633      3.182313      2.806190      1.519456      2.564898
5      1.628793      3.180431      2.783362      1.330690      2.975690
```

Application (CLI)

```
Enter choice (1-5): 2
      Art_Galleries  Dance_Clubs  Juice_Bars  Restaurants  \
Hierarchical_Cluster
1      0.843373      1.484734      2.248876      0.569882
2      0.909241      1.466329      1.340696      0.734114
3      0.825618      1.407041      0.583633      0.492210
4      0.972938      1.175141      0.463107      0.544011
5      0.904286      1.218857      1.576571      0.419571
6      0.958201      1.251511      0.381223      0.377482

      Museums  Resorts  Parks  Beaches  Theaters  \
Hierarchical_Cluster
1      1.150178  2.151124  3.192485  2.768698  1.534438
2      1.265823  2.301709  3.183797  2.851646  1.576772
3      0.915581  1.837191  3.179139  2.905318  1.855243
4      0.658644  1.577966  3.179322  2.796836  1.457740
5      0.787429  1.545714  3.180286  2.750286  1.209000
6      0.794245  1.444604  3.169496  2.853309  1.378417

      Religious_Institutions
Hierarchical_Cluster
1      2.470178
2      2.568861
3      2.882060
4      2.733616
5      3.036571
6      3.266043
```

Application (CLI)

Enter choice (1-5): 3

Hierarchical_Cluster **1** **2** **3** **4** **5** **6**

Cluster

0 0 9 121 5 0 2

1 152 23 0 1 0 0

2 0 0 35 13 4 135

3 0 12 63 140 0 2

4 13 99 25 9 1 0

5 4 15 23 9 65 0

Application (CLI)

```
Enter choice (1-5): 4
Enter values from 0 to 4 (0 = not interested, 4 = highly interested)
Art_Galleries: 2.3
Dance_Clubs: 2.1
Juice_Bars: 2.7
Restaurants: 3.2
Museums: 3.3
Resorts: 2.8
Parks: 3.2
Beaches: 1.2
Theaters: 3.5
Religious_Institutions: 0.1
-----
Assigned Cluster: 1
Cluster Label: Entertainment & Leisure Seekers
-----
```

Real World Application

- Tourism agencies can use the clusters to identify types of tourists based on preferences.
- Travel planners can recommend destinations tailored to user interests (e.g., beaches vs museums).
- Hotels and resorts can create targeted offers for specific clusters (e.g., family-friendly vs luxury).
- Marketing teams can run personalized ads for each tourist segment.
- Destination management organizations can plan resources and events based on tourist preferences.
- Local businesses can customize services according to the most common cluster in the area.

Thank You